

Beamforming and Polarization Analysis

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Abstract

Beamforming and polarization analysis are two approaches to seismic data analysis which aim to determine the wave type and source location of the incoming signal. Beamforming is viable where seismic signals are recorded by an array of one- or three-component stations, while polarization analysis techniques exist for both three-component arrays and single three-component stations. The majority of permanent seismic arrays were deployed as part of the International Monitoring System to improve the detection of nuclear explosions. Such arrays are optimized for events occurring one at a time, with an impulsive onset, tightly constrained location, and short duration. In contrast, ambient seismic noise occurs at many locations simultaneously with relatively low amplitude, has no sharp onset and may have moving sources. The range of methods that may be used to estimate the slowness and back azimuth of incoming energy arriving at the array are reviewed and discussed as well as developments of such methods that improve the observation of ambient noise, its location and character.

Single-component plane wave beamforming in the form of conventional frequency-wavenumber (f - k) analysis and data adaptive beamforming (MVDR/Capon) is introduced. Almost all array observations of seismic noise, e.g., source localization of Rayleigh, L_g , P-waves, and power measurements, can be attributed to these two techniques, which have solidified the understanding between theory, numerical modeling, and observations. The choice of array design impacts on the resolution for a given frequency range for the intended task. The theory for the underlying beamforming techniques is related to the array design and further strategies are discussed to optimize beamforming for the analysis of microseisms.

Methods for investigating the polarization of incoming ambient noise signals are presented for seismic arrays and three-component stations. Multiple ways are

reviewed to extend the beamformers described previously for an analysis of the three-component wavefield, which allows for the observation of shear surface and body waves and improves overall performance. For single station three-component sensors, a review of the most commonly used polarization methods in the context of microseism analysis is given.

2.1 Introduction

Seismologists often aim to determine the type of seismic waves present in an incoming signal and also to determine the back azimuth, which gives information on the source direction of the signal. Beamforming and polarization analysis may both be used to analyze signals from seismic events with defined onset times such as earthquakes and explosions. Ambient seismic noise, the background signal ubiquitous on all seismic records, may come from many sources simultaneously and heightened levels of seismic noise typically have no sharp onset, thus presenting additional challenges for its analysis. In this chapter, the observation and estimation of the properties of the ambient seismic wavefield are discussed. For such seismic noise investigations, beamforming analysis is the preferred approach if seismic one- or three-component array data are available and the polarization of incoming signals may be investigated using three-component arrays. If only single-station three-component data exist for a particular receiver location, then polarization analysis of these records may yield some of the desired wave properties.

2.2 Beamforming

The general idea of beamforming in seismology is to estimate the coherent portion of seismic wave energy that propagates over the seismic array and to determine its propagation characteristics (e.g., Krim and Viberg, 1996; Schweitzer et al., 2002; Rost and Thomas, 2002). This is achieved by simultaneously utilizing all sensor information of the seismic array and stacking the seismic traces according to some wave propagation model. This can be visualized with the help of a schematic example of the delay-and-sum (DAS) beamforming process (e.g., Kelly, 1967) shown in Figure 2.1. In this example, five time-shifted sensor outputs $x_i(t)$ are shown and the general form of the beam output is given as

$$y(t) = \frac{1}{M} \sum_i w_i x_i(t - \tau_i), \quad (2.1)$$

where $y(t)$ is the beam trace, M is the number of sensors, w_i are the sensor weights that differ for each type of beamformer, and τ_i are the time delays. A time shift and

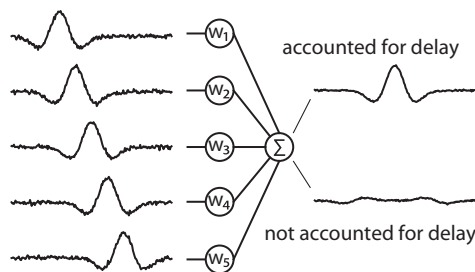


Figure 2.1. Schematic example of a delay-and-sum process on 5 synthetic traces. If the phase delay between the seismic traces is known, the summation will preserve the true signal while down weighting the instrument noise. A summation with incorrect phase delays will lead to a reduction of the true signal.

weight is applied to each sensor before the traces are combined to calculate the seismic array beam trace $y(t)$. If the true time shifts are known, one can combine all traces constructively to increase the signal-to-noise ratio and suppress individual sensor noise. The coherent summation of all sensor traces leads to the maximization of the beam power P

$$P = \int_{-\infty}^{\infty} y^2(t) dt = \int_{-\infty}^{\infty} |Y(f)|^2 df, \quad (2.2)$$

while an incoherent summation, i.e., the sensor traces are stacked with incorrect phase delays, leads to a degradation of the beam power. The right-hand side was obtained with the help of Parseval's Theorem, where $Y(f)$ is the Fourier transform of $y(t)$. The equivalent discretized equation for a beam trace of length L is given as

$$P = \frac{1}{L} \sum_{t=1}^L y^2(t) = \frac{1}{L^2} \sum_f |Y(f)|^2. \quad (2.3)$$

The normalization factor of the inverse length is important to give a measure of beam power which is independent of the seismic trace length.

The phase delays τ_i are in general unknown and have to be estimated. A common approximation is to allow only phase delays that are induced by plane waves travelling over the array in the far-field approximation, i.e., the wavefront approaches the array as a straight line without any curvature. The time delays τ_i are then derived from the knowledge of the spatial sensor locations and a wave propagation model which predicts travel times between all sensors given a slowness vector $\mathbf{s}$. As $\mathbf{s}$ is unknown, one performs a grid search over all possible slownesses to estimate the beam power P . The angle θ of the slowness vector $\mathbf{s}$

$$\mathbf{s} = s \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix} \quad (2.4)$$

which maximizes P is commonly referred to as direction of arrival (back azimuth) and s is the estimated slowness of the signal.

2.2.1 Frequency-Wavenumber Beamforming

In frequency-wavenumber (f-k) beamforming, the delay-and-sum approach is implemented in the frequency domain. It assumes plane wave propagation over the area of the seismic array and therefore assumes the seismic sources to be in the far field. From the definition of a plane wave with frequency f , imaginary number i , wavenumber vector $\mathbf{k}$, and spatial coordinate vector $\mathbf{r}$

$$x(\mathbf{k}) = e^{-i\mathbf{k}\mathbf{r}} = e^{-2\pi i f \mathbf{s}\mathbf{r}} \quad \text{with} \quad \mathbf{k} = \omega \mathbf{s} = 2\pi f \mathbf{s} \quad (2.5)$$

one can calculate the phase delay between all station pairs defined as the steering vector

$$\mathbf{a}(f, \mathbf{s}) = e^{-2\pi i f \mathbf{s}(\mathbf{r}_n - \mathbf{r}_0)}, \quad (2.6)$$

where $\mathbf{s}$ is the slowness vector of the plane wave and $\mathbf{r}_n - \mathbf{r}_0$ the distance vector between sensor n and some reference sensor at $\mathbf{r}_0$. The steering vector $\mathbf{a}(f, \mathbf{s})$ defines the phase delays induced by a plane wave propagating over the array with a slowness $\mathbf{s}$ at frequency f . These synthetic phase delays are compared to the delays induced by the observed seismic wavefield via a grid search by iterating over slowness and back azimuth of the steering vector to estimate power contributions from a variety of plane waves.

The beam trace in the frequency domain for M stations is given as

$$y(t) \xrightarrow{\mathcal{F}} Y(f) = \frac{1}{M} \sum_i w_i^* X_i(f) = \frac{1}{M} \mathbf{w}^H \mathbf{X}(f), \quad (2.7)$$

where H denotes the conjugate transpose, the superscript $*$ is the complex conjugate, $\mathcal{F}$ represents the Fourier transform of the beam trace $y(t)$ to its spectral representation $Y(f)$, and $\mathbf{X}(f)$ are the Fourier transformed seismic traces. The resulting beam power is given as

$$P(f) = \frac{1}{L^2} |Y(f)|^2 = \frac{1}{L^2 M^2} \mathbf{w}^H \mathbf{X}(f) \mathbf{X}^H(f) \mathbf{w} = \frac{1}{M^2} \mathbf{w}^H \mathbf{C}(f) \mathbf{w}. \quad (2.8)$$

The middle term

$$\mathbf{C}(f) = \frac{1}{L^2} \mathbf{X}(f) \mathbf{X}^H(f) \quad (2.9)$$

is the cross spectral density matrix (CSDM) of $M \times M$ form. For this expression the following Fourier transform convention is used

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-i2\pi f t} dt. \quad (2.10)$$

The CSDM is a Hermitian matrix that governs all auto and cross spectra information between all sensors and the diagonal terms C_{nn} form the power spectral densities $\int_{-\infty}^{\infty} C_{nn}(f)df$ of the corresponding stations (see McNamara and Boaz, 2018).

The f-k approach is formulated to maximize the beam power $\max\{P(f)\}$. Hence one wants to find the weights which maximize the last term in equation (2.8). To find the weights we reformulate the expression to

$$\mathbf{w}^H \mathbf{C}(f) \mathbf{w} = \mathbf{w}^H \mathbf{X}(f) \mathbf{X}^H(f) \mathbf{w} = |\mathbf{w}^H \mathbf{X}(f)|^2, \quad (2.11)$$

and the term simplifies to an inner product between two complex vectors (normalization is omitted in the above equation). The expression is maximized when $\mathbf{w} = \mathbf{X}(f)e^{i\phi}$, where ϕ is an arbitrary phase, given that each multiplication of the inner product results in the same complex angle (coherent phase stacking) and is summed before the calculation of the absolute value. Since one assumes plane wave propagation, the weights are set as a plane wave

$$\mathbf{w}_{fk} = \mathbf{a}(f, \mathbf{s}) \quad (2.12)$$

defined in equation (2.6). The weights are data independent and solely bound by the array geometry and the propagation characteristics of a plane wave. Hence the f-k method falls under the category of data-independent beamforming techniques. The beam power then follows from equation (2.8) as

$$P_{fk}(f, \mathbf{s}) = \frac{\mathbf{a}^H(f, \mathbf{s}) \mathbf{C}(f) \mathbf{a}(f, \mathbf{s})}{M^2}. \quad (2.13)$$

The denominator normalizes the array beam power to reflect the mean power of the signal recorded at one station and hence can differ from other normalizations which express the combined beam power (e.g., Krim and Viberg, 1996). Rewriting equation (2.13) shows the summation in detail as

$$P_{fk}(f, \mathbf{s}) = \frac{1}{M^2} \sum_n \sum_m C_{nm}(f) e^{-2\pi i f \mathbf{s}(\mathbf{r}_m - \mathbf{r}_n)}. \quad (2.14)$$

Thus, the beam power is calculated as a comparison between the cross spectrum between stations n and m and the phase delay of a synthetic plane wave governed by $f\mathbf{s}(\mathbf{r}_m - \mathbf{r}_n)$. The f-k approach is also known as Bartlett beamforming or the Bartlett processor. Identical to the time domain case, when the synthetic phase delays match the delays from the CSDM, the beam power is maximized and, in the case of a delay mismatch, the beam power degrades.

The above derivation of the beam power was carried out under the assumption of planar wave propagation, i.e., all seismic stations do not vary in elevation. For large differences in elevation and shorter signal wavelengths one is required to account

for the difference, which can be accomplished by taking into account the crustal velocity to estimate the phase delay in the vertical direction.

Alternatively, instead of correlating the observed wavefield with a synthetic plane wave, one can extend the correlation to an arbitrary wavefield configuration which leads to the class of matched field processing (Bucker, 1976; Baggeroer et al., 1993) and allows to extend beamforming to a more general form which among others can account for near field sources, heterogeneous propagation media, wavefront bending, and the sphericity of the Earth.

2.2.2 Imprint of the Array

The spatial geometry of the seismic array has a direct influence on the resolution capabilities of beamforming. A way of classifying the performance of each array is to study the beam power under the influence of a single monochromatic wave propagating over the array. The resulting power spectrum is also known as the array response function (ARF) and is defined as

$$P_{ARF}(f, \mathbf{s}) = \frac{1}{M^2} \mathbf{a}^H(f, \mathbf{s}) \mathbf{a}(f, \mathbf{s}') \mathbf{a}^H(f, \mathbf{s}') \mathbf{a}(f, \mathbf{s}), \quad (2.15)$$

where $\mathbf{C}(f) = \mathbf{a}(f, \mathbf{s}') \mathbf{a}^H(f, \mathbf{s}')$ is the CSDM of a plane wave propagating with slowness $\mathbf{s}'$. For the special case of $\mathbf{s}' = 0$, all entries in the CSDM are equal to 1, which is a popular way to display the ARF. In detail, the equation can be written as

$$P_{ARF}(f, \mathbf{s}) = \frac{1}{M^2} \sum_n \sum_m e^{-2\pi i f (\mathbf{s} - \mathbf{s}')(\mathbf{r}_m - \mathbf{r}_n)}. \quad (2.16)$$

Three ARFs are shown in Figure 2.2 for publicly available seismic arrays, i.e., (a) Gräfenberg Array, (b) Warramunga Array, and (c) Southern California Network Array. In this case $\mathbf{s}' = 0$, which represents a wave exciting all sensors simultaneously (vertical arrival). Hence the highest beam power is located at $s_x, s_y = 0$ in the power spectrum. All three arrays differ in station count, spatial configuration, and aperture, which leads to unique features in their respective ARFs.

Even though the array records only a single monochromatic plane wave, the power spectrum shows power contributions at slowness values which differ from the true slowness $\mathbf{s}'$. This is a direct result of the sparse spatial sampling (array station configuration) of the wavefield which leads to power contributions for other slownesses than $\mathbf{s}'$.

The power spectrum of an ARF can be classified into two separate areas: the beam main lobe, which is the power distribution at and in close vicinity of $\mathbf{s}'$, i.e., the true arrival direction of the signal, and the beam side lobes, which are energy contributions at other slownesses. In general, it is desired to decrease the width of

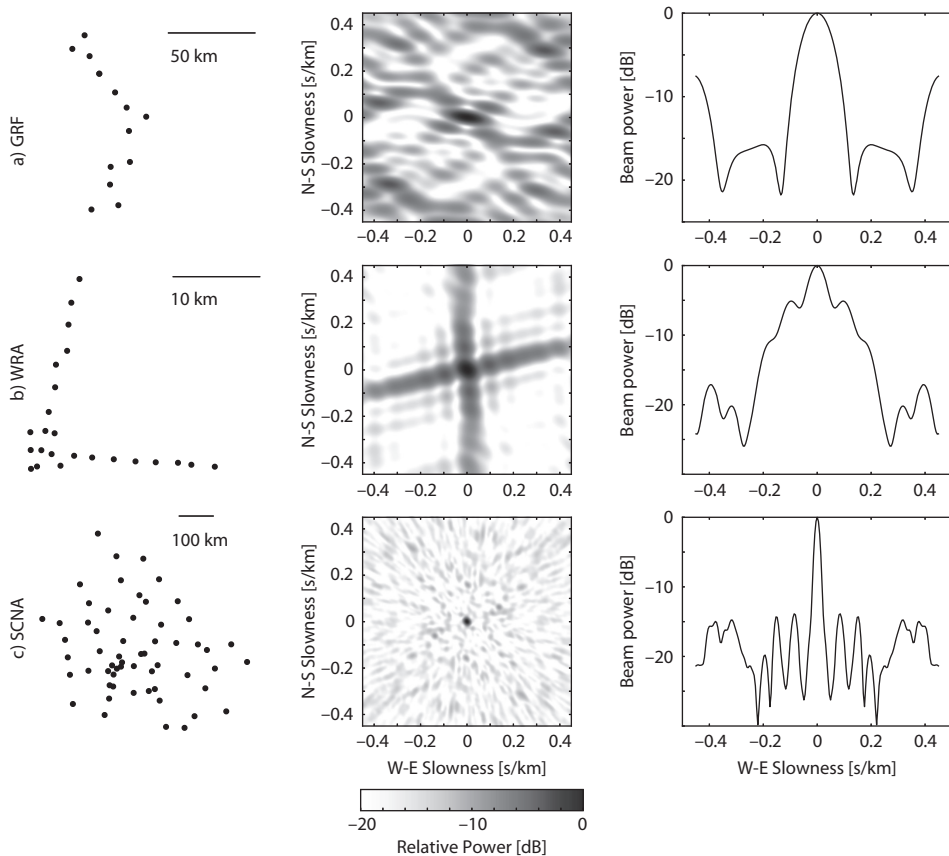


Figure 2.2. In the first column, the station configurations are displayed for (a) the Gräfenberg Array, (b) the Warramunga Array, and (c) the Southern California Network Array (not all stations). The second column displays the corresponding ARFs at $f = 0.2$, $f = 0.6$, and $f = 0.1$ Hz for GRF, WRA, and SCNA. These values are closely related to frequencies at which ambient noise studies have been performed at these arrays. The right column displays a W-E cross section of the ARFs for each array. The unique station configuration and array aperture have a direct influence on the form of the ARF.

the beam main lobe (higher resolution) and decrease the power contribution from beam side lobes. The extent of the main lobe is controlled by the geometry and aperture of the array and the frequency at which beamforming is performed.

The ARF can be seen as an elementary function of the respective array as it is the response to a plane wave. It has the unique property of shift invariance, i.e., the ARF power spectrum retains its form but is shifted in the slowness plane so that the beam main lobe is at position $\mathbf{s}'$. If multiple plane waves with differing $\mathbf{s}'$ propagate over the array, the resulting power spectrum is a superposition of ARFs

centered at the corresponding slowness vectors. Alternatively, it can be seen as a convolution between point sources at $\mathbf{s}'_i$ and the ARF. In the presence of multiple sources, beam side lobes sum constructively and can generate local maxima in the power spectrum that can be misinterpreted as real source, hence, an understanding of the ARF is important. It should be noted that, in the case of non-planar wave propagation, the form of the ARF changes.

Most of the permanent seismic arrays available globally were deployed for the purpose of nuclear signal monitoring, and hence were designed to have a high resolution for body waves above 1 Hz. Furthermore, some of these arrays were deployed to have increased resolution toward particular azimuths (e.g., the Warramunga Array) while sacrificing resolution toward others.

For the analysis of ambient noise, it is common to observe multiple arrivals of surface and body waves from a variety of azimuths, which pose differing requirements for the array geometry. It is desired to have omnidirectional resolution with minimal side lobes for an accurate estimation of multiple wave arrivals. To obtain a sharper/narrower main lobe, it is required to extend the aperture of the array, and to suppress side lobes it is required to deploy many stations which cover a variety of wave vectors $\mathbf{k}$ between all inter-station pairs. Hence, with a limited number of stations, one faces a trade-off between main lobe resolution and side lobe suppression.

2.2.3 Data Adaptive Beamforming (MVDR/Capon)

Data adaptive beamforming describes a class of beamformers that take into account the data structure to improve the power spectrum (e.g., Krim and Viberg, 1996). Unlike the data-independent f-k beamformer, whose resolution solely depends on the array configuration, the weights of data adaptive beamformers are derived from the input data.

Among the many approaches in adaptive beamforming, the minimum variance distortionless response (MVRD) or Capon beamformer (Capon, 1969) was successfully applied to ambient noise and showed superior resolution in comparison to the f-k approach (Capon, 1973). The idea is to choose the weights in a way to keep the gain constant in the array “look” direction (direction in which the array beam is focused) while suppressing contributions from all other wavenumbers. This results in a reduction of side lobes dependent on the input data. The weights of the Capon beamformer can be derived through the following minimization problem

$$\begin{aligned} \min \mathbf{w}^H \mathbf{C} \mathbf{w} \\ \text{s.t. } \mathbf{w}^H \mathbf{a}(\mathbf{s}, f) = 1. \end{aligned} \quad (2.17)$$

Hence one tries to minimize the output power of the beamformer under the constraint of maintaining the true power into the look direction. The optimization problem can be solved with the Lagrangian multiplier method

$$\mathcal{L} = \mathbf{w}^H \mathbf{C} \mathbf{w} + \Lambda (\mathbf{w}^H \mathbf{a} - 1) = \mathbf{w}^H \mathbf{C} \mathbf{w} + \Lambda (\mathbf{a}^H \mathbf{w} - 1). \quad (2.18)$$

Solving

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = 0, \quad (2.19)$$

leads to

$$\begin{aligned} \mathbf{w}^H \mathbf{C} &= -\Lambda \mathbf{a}^H \\ \mathbf{C}^H \mathbf{w} &= -\Lambda^* \mathbf{a} \\ \mathbf{w} &= -\mathbf{C}^{-1} \mathbf{a} \Lambda^*. \end{aligned} \quad (2.20)$$

Inserting the condition (2.17) and reordering yields the expression for the Lagrangian parameter

$$\Lambda^* = -\frac{1}{\mathbf{a}^H \mathbf{C}^{-1} \mathbf{a}}. \quad (2.21)$$

The weight vector follows from equation (2.20) using equation (2.21)

$$\mathbf{w}_C = \frac{\mathbf{C}^{-1} \mathbf{a}}{\mathbf{a}^H \mathbf{C}^{-1} \mathbf{a}}, \quad (2.22)$$

and leads to the Capon power spectrum

$$P_C = \mathbf{w}_C^H \mathbf{C} \mathbf{w}_C = \frac{1}{\mathbf{a}^H \mathbf{C}^{-1} \mathbf{a}}. \quad (2.23)$$

In the data adaptive case, the response of the array to a plane wave differs from the conventional f-k case. The shape of the ARF is now data dependent and shift invariance is not assured. The suppression of side lobes is best for a monochromatic plane wave and zero noise (i.e., synthetic case). Increasing the number of signals and/or noise will decrease suppression capabilities of the Capon beamformer and can, in the case of steering vector errors, result in a worse power estimation than the f-k case (e.g., Seligson, 1970). In general, the required noise levels to degrade the performance below the resolution of the f-k approach are not encountered during the analysis of ocean-induced microseisms.

The increased resolution comes at the cost of robustness (the term robustness in the beamforming jargon refers to the stability of the approach). Unlike the f-k case, the weight vector in the Capon approach is calculated via the inverse of the CSDM, which can be rank deficient in the case of limited data. Rank-deficient matrices, which necessarily have determinant equal to zero, do not have an inverse and one is forced to modify the CSDM in order to obtain an estimate of their

inverse. Successful modifications include, but are not limited to, the use of the Moore-Penrose pseudo inverse, temporal and frequency averaging, and diagonal loading (Featherstone et al., 1997).

Another robustness issue is the sensitivity to departures from the plane wave model. The Capon beamformer is subject to strong performance degradation in the presence of a propagation model mismatch (Seligson, 1970). While this is also the case for f-k, the Capon beamformer is more sensitive toward a propagation mismatch and can lead to a worse performance than the f-k approach.

2.3 Optimized Beamforming for the Analysis of Ambient Noise

The nature of the continuous background oscillations of ambient noise can be exploited in order to optimize performance and robustness of the beamforming framework. Unlike in earthquake seismology, where the source information is constrained to a relatively small time window, the signal information from ambient noise is spread over long durations and can be exploited to better estimate the resulting phase delays between all sensor pairs.

2.3.1 Pre-Processing

Quality Control – Excluding Outliers

To guarantee accurate functionality and power estimation of beamforming, a quality control of the seismic data is important. The majority of unwanted signals are either earthquakes or incoherent noise due to local sources or malfunctioned receivers. The unwanted signals become a problem once their amplitude is comparable to the amplitude of the signal. Such incoherent noise will have an imprint on the frequency spectrum and interfere with the beamforming analysis, causing the beampower to degrade. To identify seismic traces/snapshots that are perturbed by earthquake signals, a common approach is to use an event catalogue (e.g., ISC Bulletin) to identify time frames where strong earthquakes occurred and exclude them from the beamforming process. Alternatively, if accurate amplitude information is not important, one can substitute the CSDM with the spectral coherence (e.g., Capon, 1969), which normalizes the frequency spectrum (spectral whitening) and retains phase information only (discussed in more detail in *Modification of the CSDM* in this section). This is especially useful in combination with temporal averaging (Gerstoft et al., 2008) as the influence of the transient earthquake signal is averaged out. If amplitude information is of interest (e.g., Gualtieri, 2014; Farra et al., 2016), one can either identify snapshots with earthquake contributions given their differing spectral characteristics in comparison to ambient noise (Aster

et al., 2010), or one can perform a type of temporal power estimation and exclude snapshots which show power above a set threshold (e.g., Meschede et al., 2017).

Receivers that display a high, continuous, incoherent noise need to be excluded from the beamforming process.

Temporal Averaging

Given the relatively stationary nature of seismic ambient noise (within a few hours), an often deployed method is temporal averaging (e.g., Gerstoft et al., 2008). The idea is to take an array time series $\mathbf{x}(t)$ and split it into N shorter time durations of equal temporal length $\mathbf{x}_i$ with $i \in 1, \dots, N$. The shorter duration time series, also known as snapshots or temporal sub-windows, are used to construct N CSDM's which are averaged into one final CSDM

$$\mathbf{C}(f) = \frac{1}{L^2 N} \sum_i^N \mathbf{X}_i(f) \mathbf{X}_i^H(f). \quad (2.24)$$

The advantage of this averaging is that phase delays that persist over the duration of the averaging are constructively summed, while phase delays from incoherent signals are averaged out. This also means that if non-stationary signals are present, e.g., moving sources, their power will not be accurately estimated but is reduced due to the averaging procedure. Another advantage is the increase in the rank of the CSDM $\mathbf{C}$, which is especially important in the case of the Capon beamformer.

The time length, which is divided into snapshots, can reach from minutes/hours (e.g., Friedrich et al., 1998; Gerstoft et al., 2008) to multiple weeks (e.g., Euler et al., 2014) depending on the task. While temporal averaging in general increases the robustness of the beamformer, it can have a different effect on data adaptive and data-independent (f-k) beamforming. Since the Capon beamformer estimates its weight from the CSDM, its side lobe suppression capabilities are governed by the signal distribution propagating over the array. If one averages over a time duration where the wavefield is non-stationary, a variety of wavenumbers $\mathbf{k}$ populate the CSDM and reduce the suppression capabilities of the Capon beamformer.

The optimal snapshot length is dependent on the frequency of interest, the observed wavefield, and the aperture of the array. If the array aperture is large (several 100 km) and one is interested in surface waves, the snapshot length must be chosen with care given that the surface waves require a certain time to travel between all stations. This is especially important for wavefields that strongly vary with time (extreme case is surface waves from an earthquake).

In an example the importance of the snapshot length is further highlighted. Let there be a signal of interest at 0.2 Hz and our sensor has a sampling rate of 1 Hz. If one selects a snapshot length of 50 s, i.e., 10 periods of the signal are present

in one snapshot, the Fourier transformed signal consists of 26 positive valued frequency bins (including 0), which are $f_n = 0.02n$ with $n \in 0, 1, \dots, 25$, where $n = 25$ equals the Nyquist frequency. If the CSDM at 0.2 Hz is computed, one can see that additional frequencies are present, i.e., $f = 0.2 \pm 0.01$ Hz. Furthermore, one faces the issue of spectral leakage (e.g., Thomson, 1982), which is more pronounced for short snapshots. Increasing the snapshot length reduces the frequency bin width and reduces the effect of spectral leakage, and hence reduces the presence of neighboring frequencies. However, increasing the snapshot length too far (multiple hundreds to thousands of periods in one snapshot) results in a noise amplification in the phases of $\mathbf{X}(f)$, which translates into a degradation of the beamformer's performance.

With the above information one can tailor the snapshot length to a desired task, i.e., where a narrow frequency band of the wavefield is of interest one will choose a longer snapshot length, while in cases where the signal-to-noise ratio is low one might choose a shorter snapshot length.

Pre-Processing of Snapshots

Once the desired snapshot length is chosen, standard time series processing should be applied. For each snapshot the trend, mean, and sensor response should be removed and optionally a taper function should be applied. Taper functions are used to reduce finite window effects which occur during the Fourier transformation. Since each snapshot is considered as a periodic signal when the Fourier transform is applied, discontinuities at the edges are likely to appear which induce unwanted spectral information. Taper functions modify each snapshot in order to reduce certain characteristics during the Fourier transform. Commonly used taper functions include the Cosine and Hann window function to reduce spectral leakage and are more important when the snapshot length is chosen on the shorter side. One should keep in mind that taper functions directly influence the strength of signals and hence this needs to be taken into account if the signal power is of interest.

Modification of the CSDM

Overlapping snapshots is a popular method to increase the rank and robustness of the CSDM. This method is particularly useful if little data are available and one obtains a better estimated CSDM. If taper functions are used, overlapping snapshots is a great method to include the information that has been tapered off in the previous snapshot.

Another popular modification is the sensor array normalization (e.g., Capon, 1969; Haubrich and McCamy, 1969) also known as coherence. The coherence matrix is defined as

$$\hat{C}_{nm}(f) = \frac{X_n(f)X_m^H(f)}{\sqrt{|X_n(f)|^2 |X_m|^2}}. \quad (2.25)$$

This normalization retains only phase information and removes improper sensor equalization, which can occur due to path propagation effects, geographical location, faulty sensors, and gain fluctuations. The normalization becomes increasingly beneficial for seismic arrays with large apertures (e.g., Haubrich and McCamy, 1969; Gerstoft et al., 2008; Euler et al., 2014) and many stations where path propagation effects and faulty sensors due to the increased number are likely. Furthermore, it reduces the influence of earthquakes if combined with temporal averaging as their information is averaged out (e.g., Gerstoft et al., 2008).

Additionally to the above modifications it can prove beneficial to remove the diagonal elements of the CSDM, which are the auto-spectra. In the case of strong incoherent noise, the auto-spectra may result in a distorted beamforming result (e.g., Westwood, 1992; Ruigrok et al., 2017).

2.3.2 Extension to Broadband Analysis

Both the f-k and Capon beamformer as laid out in the above sections are narrowband techniques. The signal propagation characteristics are examined at one frequency f to yield the power spectrum $P(f, \mathbf{s})$. In the case of ocean-induced microseisms, where the sources are distributed in a broad frequency range, one observes only a fraction of the information present in the wavefield (see McNamara and Boaz, 2018 and Ardhuin et al., 2018). The extension to estimate the power spectrum in a broad frequency range can be accomplished through a variety of approaches (e.g., Wax et al., 1984; Wang and Kaveh, 1985; Kværna and Doornbos, 1985; Kværna and Ringdal, 1986; Westwood, 1992; Chiou and Bolt, 1993; Soares and Jesus, 2003) which can be divided into two groups. The first group focuses on combining the estimated narrowband spectra over a given frequency range in order to obtain a broadband representation. It is also known under incoherent averaged signal (IAS) method, where the incoherent averaging relates to the summation after the beam power is estimated. The second group focuses on modifying the CSDM in order to obtain the broadband representation and is also known as coherent averaged signal (CAS) method, where coherent averaging refers to methods which combine the broadband signal before the beam power is estimated.

The simplest way is to estimate narrowband spectra at multiple narrowband frequencies, and their combination results in the broadband representation

$$P_{IAS}(\mathbf{s}) = \sum_i P(f_i, \mathbf{s}). \quad (2.26)$$

Alternatively, the summation can be replaced by the median in order to suppress sharp artifacts from individual narrowband spectra. The simplicity of this method allows it to be applied over a broad frequency range. This has the additional advantage that the ARF changes with frequency, and the contribution of side lobes can average out. Another advantage is that sources present at differing frequencies do not mix with each other, which allows adaptive beamforming methods to yield better resolution. Potential disadvantages are the increased computational burden as multiple narrowband spectra need to be calculated, and stability issues in scenarios with low signal-to-noise ratio (Yoon et al., 2006).

Coherent approaches have received more attention mainly for the reason that they exploit the broadband characteristics for increased performance (e.g., Chiou and Bolt, 1993). Although most of the work has focused on the construction of a single broadband CSDM, mainly for the purpose of subspace-based beamforming (another type of adaptive beamforming), e.g., MUSIC (Schmidt, 1986), approaches that are useful for the analysis of microseisms, where multiple signals are present, are discussed.

A straightforward coherent approach is to average the CSDM $\mathbf{C}(f)$ over neighboring frequencies

$$\mathbf{C}_{CAS} = \sum_i \mathbf{C}(f_i). \quad (2.27)$$

This procedure increases the rank of the CSDM and better estimates the average phase delay between stations. A disadvantage of this method arises due to the mix of multiple frequencies in $\mathbf{C}_{CAS}$ and the subsequent projection on to a monochromatic plane wave. Hence the beamformer correlates a plane wave with a broadband signal. As the beamformer assumes that information in the CSDM is monochromatic with frequency f_p , signals that differ from f_p are shifted in slowness since $\mathbf{k} = 2\pi f\mathbf{s}$. From this linear relationship it follows that a signal that differs by $\pm 10\%$ from f_p will be estimated with a modified slowness by $\pm 10\%$ (additional information can be found in Gal et al. (2014)). Furthermore, the mixing of a frequency range is responsible for the decrease in signal coherence (Menon et al., 2014). For negligible slowness errors, this method can be used in order to increase the robustness of beamforming. The approach is similar to the short snapshot length case where wide frequency bins harbor a range of frequencies. Hence using short snapshots can lead to biased slowness estimates in the same way.

The above shortcomings can be bypassed by explicitly accounting for phase mismatch between f_n and f_p . Let there be a frequency f_n which differs from the projection frequency f_p ; one can formulate the transformation matrix $\mathbf{T}$ which transforms between two steering vectors at differing frequencies

$$\mathbf{T}(\mathbf{s}, f_p, f_n)\mathbf{a}(\mathbf{s}, f_n) = \mathbf{a}(\mathbf{s}, f_p). \quad (2.28)$$

The explicit form of $\mathbf{T}(\mathbf{s}, f_p, f_n)$ is

$$\mathbf{T}(\mathbf{s}, f_p, f_n) = \begin{pmatrix} \exp[-i2\pi \mathbf{s}\mathbf{r}_1(f_n - f_p)] & 0 & \dots & 0 \\ 0 & \exp[-i2\pi \mathbf{s}\mathbf{r}_2(f_n - f_p)] & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \exp[-i2\pi \mathbf{s}\mathbf{r}_n(f_n - f_p)] \end{pmatrix}. \quad (2.29)$$

The matrix is then used to transform the CSDM to account for the frequency mismatch

$$\mathbf{C}_{CAS}(\mathbf{s}, f_p) = \sum_n \mathbf{T}(\mathbf{s}, f_p, f_n) \mathbf{C}(f_n) \mathbf{T}^H(\mathbf{s}, f_p, f_n). \quad (2.30)$$

For $f_p = f_n$ the matrix $\mathbf{T}$ equals the identity matrix and phase information is unchanged, while for $f_p \neq f_n$ the phase information is modified. Contrary to the conventional case, here the CSDM $\mathbf{C}_{CAS}(\mathbf{s}, f_p)$ is slowness dependent, because the knowledge of the slowness is required in order to correct for the frequency mismatch. This means that in order to obtain a bias-free power spectrum, one is required to re-calculate $\mathbf{C}_{CAS}(\mathbf{s}, f_p)$ for every possible slowness $\mathbf{s}$

$$P(\mathbf{s}, f_p) = \frac{1}{M^2} \mathbf{a}^H(\mathbf{s}, f_p) \mathbf{C}_{CAS}(\mathbf{s}, f_p) \mathbf{a}(\mathbf{s}, f_p). \quad (2.31)$$

For every $\mathbf{s}$ the power contribution from a frequency other than f_p is modified by the appropriate phase delay and hence avoids slowness bias due to frequency mixing. This can be viewed as an improved version of the frequency smoothing method described in equation (2.27) for the cost of an increased computational burden. In this derivation of the method, it is assumed that the slowness remains constant over the course of the coherent summation. In cases where this assumption does not hold, bias is introduced. For the case where such slowness variations over the frequency range are known, it can be accounted for. It should be further noted that the concept of the ARF differs for this coherent approach, since signals are focused on one specific slowness at a time. While the power estimation for the projection slowness $\mathbf{s}_p$ is accurate, signals with different slowness present in the CSDM are distorted and contribute in a unique way for each slowness $\mathbf{s}_p$.

The extensions to broadband described above favor different wavefield types. For the analysis of ocean-induced microseisms, the IAS approach is mainly used (e.g., K  rna and Ringdal, 1986; Friedrich et al., 1998; Gerstoft et al., 2006, 2008; Nishida et al., 2008; Zhang et al., 2009; Reading et al., 2014; Euler et al., 2014; Gal et al., 2014, 2015) given that the signal-to-noise ratio in the band of 1–300 s

is comparably high. However, for cases with higher noise levels ($\gtrsim 1$ Hz) CAS approaches can show better performance (e.g., Yoon et al., 2006).

2.3.3 Post-Processing

In this section two methods which aim to improve the resolution of the power spectrum after its initial estimation are reviewed. Both methods fall into the category of deconvolution, where the underlying idea is to reduce the imprint of the ARF in the results. Since the ARF is constructed under the assumption of plane wave propagation, the deconvolution approach requires plane wave propagation to yield meaningful results. If seismic waves do not propagate in accordance with the plane wave model, deconvolution will introduce errors into the final results unless the true ARF is known. Hence the approaches outlined below put additional emphasis on the fulfillment of the underlying path propagation approximations.

Richardson-Lucy Deconvolution

The Richardson-Lucy (RL) deconvolution (Richardson, 1972; Lucy, 1974) was originally derived in astronomy for the purpose of deblurring images. Stars appear blurred on an image due to the telescope optics as they are convolved with a point spread function (PSF) of the instrument (similar to the concept of the ARF). With the knowledge of the instrument's PSF, the idea is to remove the imprint of the PSF from the image.

The data model is defined for an observed image $I(x)$ as

$$I(x) = B(x) * O(x), \quad (2.32)$$

where $B(x)$ is the PSF of the optical system, $O(x)$ is the true noise-free image, $*$ denotes the convolution operator, and x are the image's coordinates. It is desired to obtain $O(x)$ with the knowledge of the PSF and the observed image $I(x)$. This reconstruction can be addressed in a Bayesian framework and leads to an iterative equation for the true image

$$\hat{O}_{i+1}(x) = \hat{O}_i(x) \left[B(x)^T * \frac{I(x)}{B(x) * \hat{O}_i(x)} \right], \quad (2.33)$$

where i is the iteration number and $\hat{O}_0$ denotes the first guess for the solution, e.g., the mean of $I(x)$. It was shown that equation (2.33) deconvolves the PSF from the image $I(x)$ under a maximum likelihood constraint (Shepp and Vardi, 1982).

In cases where an object of interest is located near the edge of the image, artifacts in the form of ringing may be introduced during the iteration procedure (Gibbs oscillations). The artifacts can be reduced by an extension proposed by Bertero and Boccacci (2005), which modifies the original equation to

$$\bar{O}_{i+1}(x) = \bar{w}(x) \bar{O}_i(x) \left[\bar{B}(x)^T * \frac{\bar{I}(x)}{\bar{B}(x) * \bar{O}_i(x) + b} \right]. \quad (2.34)$$

The objects denoted with a bar have been extended by zero padding, b is a regularization parameter that equals the background noise, and $\bar{w}(x)$ is a mask object, which defines the area of influence of the PSF. An in-depth explanation of each object and how it can be merged with f-k beamforming is given by Gal et al. (2016).

The RL deconvolution can be directly transferred and applied for the case of f-k beamforming, where the power spectrum is a convolution between true source location in the wavenumber plane and the ARF. The first successful application of the RL deconvolution in the observation of microseisms was demonstrated by Nishida et al. (2008) with the Hi-net array in Japan, where smearing of the f-k power spectrum induced by Hi-net's elongated station configuration was reduced. Other studies have since then evaluated the performance of the RL deconvolution approach on ambient noise beamforming results in more detail (e.g., Picozzi et al., 2010; Gal et al., 2016).

The maximum likelihood constraint makes the RL deconvolution method very robust, but one feature in particular that differs between seismology and astronomy should be highlighted. PSFs in astronomy generally exhibit small side lobes (small in comparison to seismic ARFs), which decay with an increasing distance from the main lobe. In seismology, the ARF does not decay in this way and side lobes extend into infinity. This can pose an ambiguity in the choice of the mask $\bar{w}(x)$ (Gal et al., 2016), which has a direct influence on the final solution. Hence testing of the $\bar{w}(x)$ for each array configuration is required for an optimal solution. Additionally, the optimal stop criterion is unknown and needs to be selected by the user.

An example demonstrating the capabilities of the RL deconvolution is shown in Figure 2.3. The ARF, f-k, and Capon beam power for a 1-hour-long recording with the Gräfenberg Array on 1 January 2017 00:00:00 UTC is displayed in Figure 2.3a–c. The analysis is carried out at 0.067 Hz (15 s) and the f-k results show one broad main lobe. The Capon beamformer shows improved resolution and can separate two sources which appear as one in the f-k result. The RL deconvolution result (Figure 2.3d), here shown after 50 iterations, shows a move of the main lobe toward one of the maxima found with the Capon beamformer but does not identify the second source to the north.

CLEAN

The CLEAN algorithm, initially developed in radio astronomy (Högbom, 1974), follows the same principle, i.e., the deconvolution of the ARF from an image. The idea is to locate the brightest spot in the image $I(x)$ and remove a fraction of the PSF $P(x)$ associated with the brightness maximum

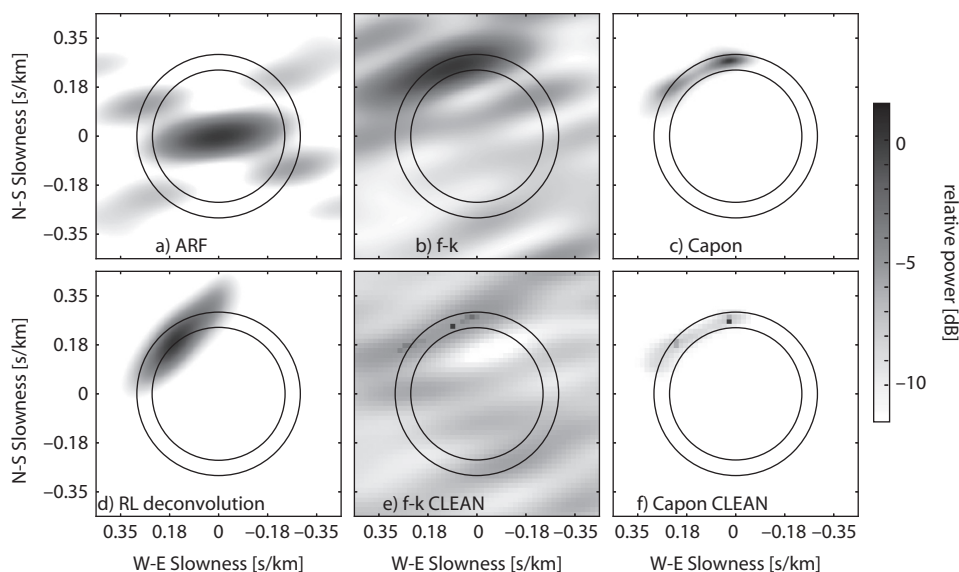


Figure 2.3. Deconvolution approaches demonstrated with the Gräfenberg Array on 1 January 2017 00:00:00 UTC at 0.067 Hz. The top row shows the (a) ARF, (b) f-k, and (c) Capon beam power. The bottom row shows (d) the RL deconvolution calculated from the f-k result, (e) the CLEAN algorithm in combination with f-k beamforming, and (f) the CLEAN in combination with the Capon beamformer. The two black circles of constant velocity are set at 3 and 4 km/s.

$$\hat{I}^{i+1}(x) = I^i(x) - \phi I^i(x_{max})P(x_{max}) \longrightarrow \hat{O}(x) = \sum_i \phi I^i(x)P(x_{max}), \quad (2.35)$$

where the superscript i denotes the iteration number, ϕ is the CLEAN control parameter which regulates the strength of the removed PSF, and x_{max} points toward the maximum in the image. In the case of $P(x_{max})$, the main lobe of the PSF is centered at x_{max} . Hence one iteratively removes a fraction of the brightness maximum and its associated side lobes from $I^i(x)$. Since the power and location of the removed brightness are known, one can combine the removed brightness to form a representation $\hat{O}(x)$ free of side lobes, which is shown in the right-hand side of equation (2.35).

In its original form, CLEAN performs the deconvolution procedure on the observed image and hence the algorithm is directly transferable to images from f-k beamforming. In the case of beamforming, an alternative approach for the implementation of CLEAN is possible, where the cleaning procedure is directly applied to the CSDM (Wang et al., 2004; Sijtsma, 2007). This has the advantage that CLEAN can be applied to data adaptive beamformers (Gal et al., 2017), which have a data-dependent ARF.

The underlying equation is

$$\mathbf{C}^{i+1}(f) = \mathbf{C}^i(f) - \phi P^i(f, \mathbf{s}_{max}) \mathbf{a}(f, \mathbf{s}_{max}) \mathbf{a}^H(f, \mathbf{s}_{max}), \quad (2.36)$$

where $\mathbf{s}_{max}$ is the slowness vector which maximizes P^i . Hence this implementation removes a fraction of the power associated with the strongest signal in P^i . The term $\mathbf{a}\mathbf{a}^H$ generates the appropriate phase delays which need to be removed. This can be alternatively seen as an introduction of a synthetic point source of negative power at the position of the dominant source. In analogy to the original CLEAN, since the removed power is known, one can combine all “cleaned” point sources and combine them to yield the side lobe free power spectrum

$$P_{CLEAN}(f, \mathbf{s}) = \sum_i \phi P^i(f, \mathbf{s}_{max}). \quad (2.37)$$

A range of stopping criteria can be applied to CLEAN, ranging from power thresholds, to total power decomposition, to user-defined criteria. Regardless of the choice, the iteration procedure should be stopped once an eigenvalue of the CSDM becomes negative. A negative eigenvalue would suggest negative energy present in the CSDM which is nonphysical.

An example for the CLEAN approach is shown in Figure 2.3e,f for the f-k and Capon algorithm. In the f-k case, after CLEAN has removed a major part of the main lobe’s beam power, the single maximal peak separates into its two sources. Given that a major part of the power was removed with an incorrect slowness vector, the successive iterations try to compensate for it, which can lead to inaccurate source identification (documented in Gal et al. (2016)). CLEAN in combination with the Capon beamformer (Figure 2.3f) shows highest resolution.

2.3.4 Array Design and Use

The analysis of microseisms by means of beamforming benefits most from arrays with narrow beam main lobes and a strong side lobe suppression. The width of the main lobe is narrowed by increasing the aperture of the array, while the side lobes are more strongly suppressed when as many differing wavenumbers as possible are realized by inter-station pairs. The second point is especially important given that multiple sources from differing azimuths are likely to be present in the estimate wavefield. One could naively assume that increasing the amount of stations in an array automatically results in a better side lobe suppression, but this is only the case if the additional stations are positioned in a way which covers inter-station distances that were not realized by the previous configuration. An example is the regular grid array, which if extended periodically will not improve the side suppression.

Optimal Array

An optimal array configuration for the analysis of seismic ambient noise is realized by the spiral shape (e.g., Kennett et al., 2015). A synthetic example of a spiral array configuration with 101 sensors and an aperture of 220 km is shown in Figure 2.4a. The ARF for this configuration at 0.15 Hz is displayed in Figure 2.4b. The advantage of the spiral configuration is the coverage of a range of inter-station distances, i.e., the sampling of a variety of wavenumber vectors.

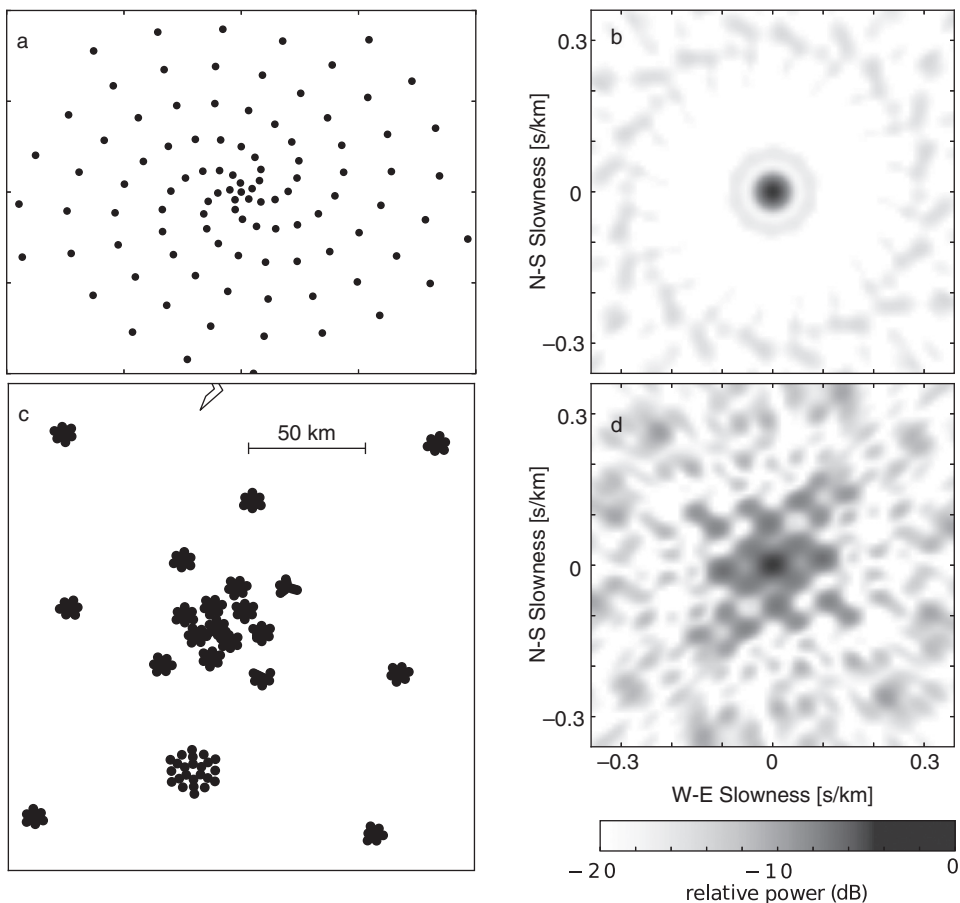


Figure 2.4. First row displays a synthetic spiral array configuration with 101 stations and the corresponding ARF at $f = 0.15$ Hz. The second row displays the Large Aperture Seismic Array configuration with over 300 stations and its corresponding ARF at the same frequency. The aperture of the synthetic array was chosen to match the one of LASA. The spiral array has a better coverage over the wavenumber space, which allows it to strongly suppress the beam side lobes, which is preferable for the analysis of microseisms.

Furthermore, the configuration is an optimal trade-off between amount of stations and side lobe suppression and shows good performance over a range of frequencies with omnidirectional capabilities. This can be visualized when compared with the large aperture seismic array (LASA), which has 300+ stations (Figure 2.4c). The ARF of LASA at 0.15 Hz is shown in Figure 2.4d and displays stronger side lobes than the synthetic array with 1/3 of LASA's stations. For higher frequencies, which LASA was initially designed for, the side lobes still extend 0.1 s/km from the main lobe and are suppressed below -20 dB for higher slownesses. The synthetic array configuration on the other hand shows sparse side lobe maxima with power levels below -15 dB for higher slownesses at 1 Hz, which could be further suppressed if the amount of stations would approach the LASA station count.

Imperfect Arrays

In most cases, the optimal array in the desired location is unavailable and one has to work with available arrays which deviate from the optimal form. This situation occurs with grid-shaped temporary arrays which are commonly deployed for tomography purposes where a constant spatial sampling of the wavefield is desired. For such arrays, the ARF side lobes are large and can strongly influence the beam-forming results. To mitigate the strong side lobes, one can adjust station weights to obtain an improved ARF.

The above idea is discussed using a temporary array deployed in Tasmania, Australia (station configuration - Figure 2.5a). The SETA array comprises 42 stations in a relatively grid-like configuration, and its ARF is shown in Figure 2.5b at 0.2 Hz. Six strong side lobes are visible with power levels up to ~ -3 dB, which are the reason for strong artifacts when estimating the ambient noise power spectrum. Instead of using the conventional ARF equation, one can extend it to include station weights (Lacoss et al., 1969), which leads to

$$P_{ARF}(f, \mathbf{s}) = \frac{1}{M^2} \sum_n \sum_m w_n w_m e^{-2\pi i f (\mathbf{s} - \mathbf{s}')(\mathbf{r}_m - \mathbf{r}_n)}. \quad (2.38)$$

The station weights w_n are set to 1 for the conventional case for which the above equation equals equation (2.14). The idea is to search for stations which are positioned at locations not conforming to the regular grid, i.e., stations which sample different wavenumbers, and increase their weight (gain). If this is done for the SETA array, one obtains the modified ARF displayed in Figure 2.5c. The weights of stations MOO and TAU (Figure 2.5a) were empirically increased, which results in a side lobe reduction to ~ -5.5 dB. In this particular example, the weight value of station MOO and TAU was changed from 1 to 4.

The modification of the weights comes at a trade-off, as can be seen in Figure 2.5c where new side lobes appear in contrast to the unweighted case. By

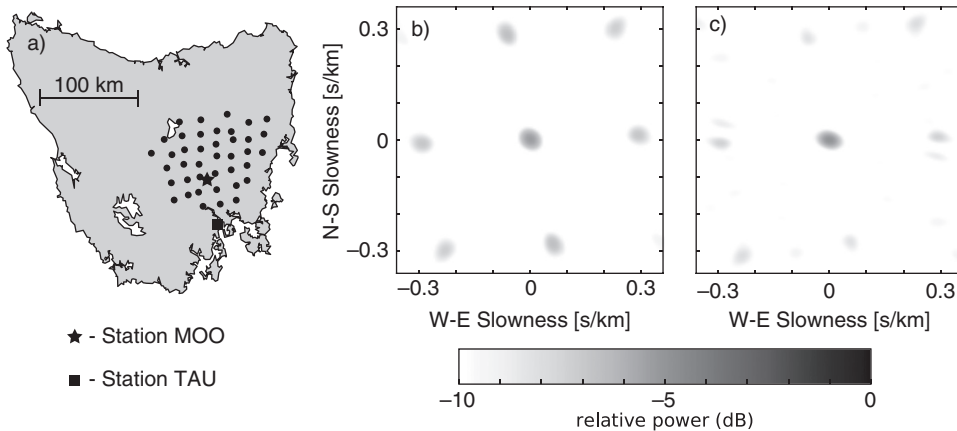


Figure 2.5. Displayed is a temporary seismic array (black dots) deployed in Tasmania, Australia (a), with an approximately mesh-like station configuration. Additionally, two permanent stations are displayed (MOO and TAU) in the area of the temporary array. Mesh-like array configurations suffer from a limited sampling of the wavenumber space and (b) result in strong side lobes visible in the ARF ($f = 0.2$ Hz). By increasing the importance (station weight) of the two permanent stations, one boosts the contribution of underrepresented wavenumbers in the beamformer process, which results in an overall decrease of side lobe power levels. The modification of station weights comes at a trade off and generates new side lobes, but under careful optimization a more suitable ARF can be achieved.

suppressing the large side lobes via the station weight modification, one increases the power of other features in the power spectrum. In this case, the modified ARF is preferred over the original case as side lobe power levels are much smaller. The station weights are applied in the same way for the desired beamformer.

2.4 Three-Component Beamforming

Three-component plane wave beamforming, which is able to estimate plane wave propagation on all three components (i.e., including Love, SV, SH, and L_g waves), is constrained by the same approximations as the one-component case. Hence the estimated signal propagates as a plane wave over all three components in the far-field approximations. Two groups of extensions in which three-component beamforming can be realized are discussed. The first group reduces the two horizontal components to a single dimension by rotating the horizontal components into the desired look direction to separate radial and transverse signals (e.g., Toksöz and Lacoss, 1968; Lacoss et al., 1969; Haubrich and McCamy, 1969; Capon, 1973; Kväerna and Doornbos, 1985; Friedrich et al., 1998; Nishida et al., 2008; Brooks et al., 2009; Poggi and Fäh, 2010; Gibbons et al., 2011; Behr et al., 2013)

and applies one-component beamforming separately on each component. The second group makes use of all three components simultaneously (e.g., Esmersey et al., 1985; Wagner, 1996; Wagner and Owens, 1996; Wagner, 1997; Miron et al., 2005; Riahi et al., 2013; Gal et al., 2016; Juretzek and Hadziioannou, 2016) and will be discussed on the basis of two potential implementations in the following section.

2.4.1 Dimensional Reduction

The reduction from two horizontal components to a single dimension may be achieved with

$$\begin{aligned}\mathbf{x}_R(t, \theta) &= \mathbf{x}_N(t) \cos \theta + \mathbf{x}_E(t) \sin \theta \\ \mathbf{x}_T(t, \theta) &= -\mathbf{x}_N(t) \sin \theta + \mathbf{x}_E(t) \cos \theta,\end{aligned}\tag{2.39}$$

where the subscripts N, E, R, and T denote the north, east, radial, and transverse components, respectively, and θ is the look direction (back azimuth) of the array. In analogy to the one-component case, the rotated radial and transverse components can be used as an input to any one-component beamformer to generate the respective CSDM

$$\begin{aligned}\mathbf{C}_R(f, \theta) &= \frac{1}{L^2} \mathbf{X}_R(f, \theta) \mathbf{X}_R^H(f, \theta) \\ \mathbf{C}_T(f, \theta) &= \frac{1}{L^2} \mathbf{X}_T(f, \theta) \mathbf{X}_T^H(f, \theta)\end{aligned}\tag{2.40}$$

and the radial and transverse power spectra follow as

$$\begin{aligned}P_R(f, \mathbf{s}) &= \frac{1}{M^2} \mathbf{a}^H(f, \mathbf{s}) \mathbf{C}_R(f, \theta) \mathbf{a}(f, \mathbf{s}) \\ P_T(f, \mathbf{s}) &= \frac{1}{M^2} \mathbf{a}^H(f, \mathbf{s}) \mathbf{C}_T(f, \theta) \mathbf{a}(f, \mathbf{s}).\end{aligned}\tag{2.41}$$

This approach has the advantage that all one-component beamforming extensions can be directly applied to the horizontal components and the simplicity and robustness remain. A disadvantage is that the intrinsic separation of the sensor data (equation 2.39) is based on the assumption of straight propagation paths and might not hold for large arrays where wavefront bending can occur. Assuming the ray path deviates from its expected direction by 5.7° , 10% signal leakage is introduced from the transverse to the radial component (based on a simple trigonometric calculation) and *vice versa*. Hence, small deviations from the plane wave model in the far field induce a mixing of radial and transverse energy. Additionally, anisotropy of the elastic modulus can also lead to deviations between polarization and propagation direction.

2.4.2 Polarization Beamforming

Polarization beamforming utilizes all three components simultaneously by exploiting the polarization characteristics of the propagating waves. The following two approaches, which employ three-component polarization beamforming, are summarized. The method of Esmersey et al. (1985), recently applied to ambient noise (e.g., Riahi et al., 2013; Juretzek and Hadziioannou, 2016), extends the Fourier transformed data vector to

$$\mathbf{X}_{3C}(f) = [X_{E,1}(f), \dots, X_{E,M}(f), X_{N,1}(f), \dots, X_{N,M}(f), X_{Z,1}(f), \dots, X_{Z,M}(f)], \quad (2.42)$$

where X_{nm} denotes the Fourier transform of the n th component and m th sensor. The three components are concatenated, where the order of the components is interchangeable, but is required to be consistent for the successive steps. The three-component CSDM leads to

$$\mathbf{C}_{3C}(f) = \frac{1}{L^2} \mathbf{X}_{3C}(f) \mathbf{X}_{3C}^H(f) \quad (2.43)$$

and is a $3M \times 3M$ matrix. A more intuitive form of the 3C CSDM is (dependence on f is omitted)

$$\mathbf{C}_{3C} = \begin{bmatrix} \mathbf{C}_{EE^*} & \mathbf{C}_{EN^*} & \mathbf{C}_{EZ^*} \\ \mathbf{C}_{NE^*} & \mathbf{C}_{NN^*} & \mathbf{C}_{NZ^*} \\ \mathbf{C}_{ZE^*} & \mathbf{C}_{ZN^*} & \mathbf{C}_{ZZ^*} \end{bmatrix}. \quad (2.44)$$

$\mathbf{C}_{3C}$ is a Hermitian matrix as in the one-component case and governs phase delay information between all stations and components.

To estimate the power of the desired polarization it is required to define a propagation model which describes the particle motion of the signal. Esmersey et al. (1985) obtained the representation via the following equation

$$\hat{\mathbf{a}}(f, \mathbf{s}, \xi) = \mathbf{c}(\xi) \otimes \mathbf{a}(f, \mathbf{s}), \quad (2.45)$$

where $\mathbf{c}(\xi) = [c_E(\xi), c_N(\xi), c_Z(\xi)]^T$ is the normalized complex polarization state (Samson, 1983) with the polarization parameter ξ and $\otimes$ is the Kronecker product. Hence the one-component steering vector is extended to a three-component propagation times the polarization state of the desired signal. The three-component power spectrum for a particular polarization is then

$$P_{3C}(\mathbf{s}, f, \xi) = \frac{1}{M^2} \hat{\mathbf{a}}^H(f, \mathbf{s}, \xi) \mathbf{C}_{3C}(f) \hat{\mathbf{a}}(f, \mathbf{s}, \xi). \quad (2.46)$$

Hence the approach uses all stations and components of an array to evaluate the power contribution of a plane wave with the polarization state $\mathbf{c}(\xi)$. To obtain the

complete power representation of all signals present in the wavefield, it is required to sum over all possible polarization states.

Another three-component beamforming framework which was recently applied to microseism analysis (Gal et al., 2016; Liu et al., 2016; Gal et al., 2017) was given by Wagner (1996), where a steering vector matrix is defined as

$$\mathbf{e}(f, \mathbf{s}) = [\mathbf{a}_E(f, \mathbf{s}), \mathbf{a}_N(f, \mathbf{s}), \mathbf{a}_Z(f, \mathbf{s})], \quad (2.47)$$

which in more detail is given as

$$\mathbf{e}(f, \mathbf{s}) = \begin{bmatrix} a_{E1} & \dots & a_{EM} & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \dots & 0 & a_{N1} & \dots & a_{NM} & 0 & \dots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 & a_{Z1} & \dots & a_{ZM} \end{bmatrix}^T. \quad (2.48)$$

In accordance to the previous approach, the components are interchangeable but are required to be consistent for all remaining steps. Here a_{nm} are the conventional steering elements of $\mathbf{a}(f, \mathbf{s})$, where n denotes the component and m the station. Hence all components are populated with the propagation of a plane wave. The steering matrix is then used to obtain the 3×3 polarization covariance matrix

$$\mathbf{Y}_{3C}(f, \mathbf{s}) = \frac{1}{M^2} \mathbf{e}^H(f, \mathbf{s}) \mathbf{C}_{3C}(f) \mathbf{e}(f, \mathbf{s}). \quad (2.49)$$

This projection and the resulting polarization covariance matrix can be viewed as the sum of all signals that propagate over the array as plane waves with a coherent polarization. This is in contrast to the previous approach, where a projection on a single polarization state is used, while here all coherent polarizations between all stations are retained.

Single-station polarization processing (e.g., Vidale, 1986) is applied to obtain the power spectrum by solving the eigenvalue problem for $\mathbf{Y}_{3C}$

$$\mathbf{Y}_{3C}(f, \mathbf{s}) = \mathbf{u}(f, \mathbf{s}) \boldsymbol{\lambda}(f, \mathbf{s}) \mathbb{I} \mathbf{u}^H(f, \mathbf{s}), \quad (2.50)$$

where the unitary matrix $\mathbf{u}(f, \mathbf{s})$ comprises the eigenvectors, $\boldsymbol{\lambda}(f, \mathbf{s})$ are the eigenvalues, and $\mathbb{I}$ is the identity matrix (for readers unfamiliar with the principal component approach, more details can be found in section 2.5.1). The eigen decomposition parametrizes the polarization states into an orthogonal basis (e.g., Samson, 1983; Wagner, 1996), where the eigenvalues denote the contribution of the polarization direction given by its eigenvector (for more detail see section 2.5). The total power spectrum (power from all 3 components) is obtained from the eigenvalues

$$P_{3C}(f, \mathbf{s}) = \sum_{i=0}^2 \lambda_i(f, \mathbf{s}). \quad (2.51)$$

Although not specifically stated in Wagner (1996), if the average signal power per station is desired the normalization of $\frac{1}{M^2}$ is required for the f- k_{3C} approach. The eigenvector of each polarization direction carries the detailed power contribution from each component, e.g., the eigenvector corresponding to λ_0 (the strongest polarization estimated by the principal component analysis) is

$$\begin{bmatrix} \mathcal{A}_E \\ \mathcal{A}_N \\ \mathcal{A}_Z \end{bmatrix} = |\mathbf{u}_0(\mathbf{s}, f)| = \begin{bmatrix} |\psi_E| \\ |\psi_N| \\ |\psi_Z| \end{bmatrix}. \quad (2.52)$$

The complex numbers ψ_i are the individual contributions to the polarization for each component. The power contribution of each component is obtained as the square of the individual amplitudes $\mathcal{A}$ times the corresponding eigenvalue

$$\begin{bmatrix} P_E \\ P_N \\ P_Z \end{bmatrix} = \begin{bmatrix} \mathcal{A}_E^2 \\ \mathcal{A}_N^2 \\ \mathcal{A}_Z^2 \end{bmatrix} \lambda_0. \quad (2.53)$$

To rotate the components into radial and transverse, one can use the rotation matrix $\tilde{\mathbf{M}}$

$$\tilde{\mathbf{M}} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \quad (2.54)$$

to translate the horizontal components of the eigenvector $\psi_{R,T} = \tilde{\mathbf{M}}\psi_{N,E}$, where θ is the angle between north and the radial component, i.e., the back azimuth. After the rotation, inserting into equation (2.52) followed by (2.53) will lead to the beam power for the radial P_R and transverse P_T component respectively. In general only the two largest eigenvalues are considered, as the third is usually dominated by noise (Gal et al., 2016). Adaptive beamformers such as MVDR (Capon, 1969), MUSIC (Schmidt, 1986), and others can be directly implemented into the three-component approaches presented above.

The advantage of polarization three-component beamforming is the increased resolution of the power spectrum given that all three components of a station are simultaneously evaluated for coherent energy propagation. This also results in a higher sensitivity of the approach toward wavefront bending, as the polarization direction at each station is assumed to be the same. If this is not the case, the beam power will start to degrade. While this could be used as a quality control to test if the underlying assumptions are fulfilled, it limits the range at which polarization beamforming is the preferred approach. In the case of wavefront bending, the approach via dimensional reduction might still give meaningful results under the consideration of signal leakage from orthogonal horizontal components.

2.5 Polarization Analysis

Single station polarization analysis is a useful tool for the study of the ambient noise field where the approximation and/or conditions for beamforming cannot be fulfilled or where array data are not available. In such a case, a single seismic station with a three-component sensor can be used to estimate the properties of the seismic wavefield. The estimated parameters of interest are generally the back azimuth of the polarization direction, rectilinearity/planarity, angle of incidence, and phase relationships between the principal axes of particle motion. In the following section, the commonly used methods and the required steps to ensure an accurate analysis of the ambient noise are introduced.

2.5.1 The Principal Component Analysis Approach

A popular approach is the estimation of the particle motion via the principal component analysis of the data covariance matrix (e.g., Flinn, 1965; Vidale, 1986; Jurkevics, 1988; Bataille and Chiu, 1991; Schulte-Pelkum et al., 2004) or the cross spectral density matrix (e.g., Samson, 1983; Park et al., 1987; Koper and Hawley, 2010; Schimmel et al., 2011). The approaches are similar for the time, frequency, and time-frequency (e.g., S-transform or wavelet transform) domain when following the principal component analysis approach. The underlying principle on the case of the time domain is demonstrated here. For a three-component sensor

$$\mathbf{x}(t) = [x_E(t), x_N(t), x_Z(t)]^T \quad (2.55)$$

the covariance matrix is a positive semidefinite 3×3 matrix

$$\mathbf{Y} = \frac{1}{L} \sum_{t=1}^L \mathbf{x}(t) \mathbf{x}^T(t), \quad (2.56)$$

where the superscript T is the transpose operator. The diagonal elements are the variances of each component, while the off-diagonal elements are a measure of correlation between the two respective components. The matrix is used to estimate the phase relationship between the three components of a single station. Note, in the case of complex data, e.g., the analytical signal in time domain or complex data in frequency domain, the transpose operator T is replaced by the Hermitian operator H . In the time domain, pre-filtering of the seismic data to the desired frequency range is required followed by an averaging of the covariance matrix over a time window with length N , where the minimal length of the time window should allow the presence of a full period of the longest wavelength present in the frequency band of interest.

To obtain estimates of the particle motion of the wavefield, the covariance matrix is decomposed using principal component analysis (PCA). In this case, either the eigenvalue or singular value decomposition can be used since the matrix is positive semidefinite. Through the PCA, the covariance matrix is transformed into an orthonormal basis which best describes the underlying structure of the recorded data. In the case of a single seismic wave recorded by a three-component sensor, the PCA transforms the data into the principal axes of the particle motion. For multiple and/or non-stationary signals, where the covariance matrix is governed by a mixture of particle motions, the decomposition via the PCA is not able to decompose the signals into its unique particle motions and the resulting orthonormal basis can be an abstract one. For an accurate separation of multiple polarization states, multiple stations are required (e.g., Roueff et al., 2009).

Assuming an accurately decomposed polarization state, it is common practice to sort the eigenvalues and their corresponding eigenvectors so that $\lambda_0 > \lambda_1 > \lambda_2$. The eigenvalues describe the contribution of the principal axes (eigenvectors) of the estimated signal's particle motion (e.g., Samson, 1983; Park et al., 1987), and hence represent power along the principal axes. The computation of various ratios of eigenvalues λ or eigenvectors $\mathbf{u}$ of $\mathbf{Y}$ is required to infer polarization and other properties. The direction of polarization is estimated from the horizontal components of the dominant polarization direction by utilizing the information in the eigenvector

$$\mathbf{u}_0 = [\psi_E, \psi_N, \psi_Z]^T \quad (2.57)$$

$$\Theta = \arctan \left(\frac{\psi_E}{\psi_N} \right), \quad (2.58)$$

which in the case of P , SV , and Rayleigh waves coincides with the back azimuth $\pm 180^\circ$. The $\pm 180^\circ$ ambiguity arises since a single three-component station cannot estimate the propagation direction of a seismic wave from the polarization direction alone, but requires an additional constraint. For body waves one can make use of the dip of the direction of polarization

$$\Theta_V = \arctan \left(\frac{\psi_Z}{\sqrt{\psi_N^2 + \psi_E^2}} \right) \quad (2.59)$$

while Rayleigh waves are usually assumed to be retrograde and their propagation direction is estimated by the phase delay between the vertical and horizontal phase angles. Care should be taken when estimating polarization parameters via the frequency domain, where the polarization ellipsoid is prone to “tip over” and switch between prograde and retrograde (Koper and Hawley, 2010).

Furthermore, the rectilinearity

$$G = 1 - \frac{\lambda_1 + \lambda_2}{2\lambda_0}, \quad (2.60)$$

and planarity

$$\mathcal{P} = 1 - \left(\frac{2\lambda_2}{\lambda_0 + \lambda_1} \right) \quad (2.61)$$

of the particle motion state can be defined (Jurkevics, 1988). For $\lambda_0 \gg \lambda_{1,2}$, the polarization is rectilinear (i.e., a pure body wave or Love wave) and is solely parametrized by a single polarization direction. For $\lambda_{0,1} \gg \lambda_2$, the particle motion is parametrized by the two largest eigenvalues and confined in a plane as is the case for a Rayleigh wave.

The approaches in the time-frequency and frequency domain follows the same logic, where the spectral covariance matrix is calculated from $\mathbf{X}(t, f)$ or $\mathbf{X}(f)$, respectively. One difference is that the resulting eigenvectors are complex and to apply the above equations the real part of the eigenvector needs to be used.

2.5.2 Instantaneous Polarization Attributes

The introduction of the analytic signal in seismic signal processing can be used to estimate the instantaneous frequency (Taner et al., 1979) and polarization attributes (Vidale, 1986; Schimmel and Gallart, 2003). The Hilbert transform $\mathcal{H}(x)$, which is used to obtain the phase-shifted function of x by $\frac{\pi}{2}$, is applied to the observed data $x(t)$ and added as the imaginary part to yield the analytic signal representation

$$\tilde{\mathbf{x}}(t) = [x_E(t) + i\mathcal{H}(x_E(t)), x_N(t) + i\mathcal{H}(x_N(t)), x_Z(t) + i\mathcal{H}(x_Z(t))]^T. \quad (2.62)$$

Following the covariance approach, the eigen analysis yields complex eigenvectors which contain the phase information (Vidale, 1986). To obtain the back azimuth of the polarization direction, only the real part of the eigenvector in equation (2.58) is used (as for the time-frequency and the frequency domain). This approach has the advantage that the covariance matrix is not required to be averaged over a portion of the desired wavelength (as is the case in the real valued analysis) and hence allows to estimate the polarization attributes at a given time $\tilde{\mathbf{x}}(t)$. This is possible owing to the Hilbert transform, which is computed on the whole seismic record, i.e., it uses signal information from the whole data window to compute the phase-transformed signal at a time t .

2.5.3 Ambient Noise Polarization Analysis

The nature of the ambient noise field, where multiple incoherent signals are likely to be present at one time, means that the estimation of the polarization is not trivial.

Whether the time, the time-frequency, or the frequency domain approach is more advantageous depends on the properties of the wavefield and the desired task. The frequency domain approach is advantageous if multiple incoherent signals can be separated in the frequency domain, i.e., unique sources occupy different frequency bands. If multiple incoherent signals are separated in time but share the same frequency band, the time or the time-frequency domain approach in combination with the analytic signal is preferable. The time-frequency domain approach combines strengths from both and gives the user more control over the polarization analysis.

All three of the above approaches have been successfully used for the analysis of the seismic ambient noise wavefield. It was shown that in the case of a relatively stationary source (stationary within the length of a data window) the use of long time windows in time and frequency domain can result in stable estimates (e.g., Schulte-Pelkum et al., 2004; Tanimoto et al., 2006; Koper and Hawley, 2010). These studies focused on the secondary microseism peak, which has an increased likelihood of encountering a single strong source compared to even higher frequency bands where polarization analysis with long time windows is unlikely to succeed unless additional constraints such as degree of polarization are employed (e.g., Koper and Burlacu, 2015) or a large number of observations is used to increase confidence.

For non-stationary sources, a better time resolution is required which can be achieved by transforming into the time-frequency domain (e.g., Schimmel et al., 2011). The transformation has the advantage that optimal frequency-dependent windows are used to sample the seismic data, which can take into account the non-stationary evolution of the wavefield.

When analyzing seismic ambient noise data, strategies to increase robustness should be employed regardless of the approach and domain chosen. The first is to use a large sample size of observations to increase robustness of the analysis as statistical outliers are down weighted. Another option is the inclusion of a broader frequency range for the computation of the particle motion, which, as long as the broadband signal has the same spatial origin, will result in a more robust estimation of the particle motion (e.g., Tanimoto et al., 2006). If, on the other hand, multiple sources are likely to enter the polarization analysis due to an extended frequency range, the robustness of the observations is likely to decrease.

Another strategy is to employ a framework which can estimate the purity of an estimated polarization state. This can be estimated for long time windows as a comparison between eigenvalues (Samson, 1983; Koper and Hawley, 2010; Koper and Burlacu, 2015) or for instantaneous polarization attributes with a degree of

polarization measure (e.g., Schimmel and Gallart, 2003, 2004; Schimmel et al., 2011; Sergeant et al., 2013; Davy et al., 2015).

2.5.4 Beamforming versus Polarization Analysis on Observed Data

A convenient way to test the polarization analysis is with the help of a three-component array where the results can be verified via beamforming. Additionally, it provides multiple stations which should observe the same sources, and hence show similar particle motions across all stations. Use is made of the Pilbara Array in Australia and the Gräfenberg Array in Germany with 13 three-component stations each.

The first example with the Pilbara Array details observations for two differing frequency bands centered on $f_1 = 0.17 \pm 0.1$ Hz and $f_2 = 0.35 \pm 0.2$ Hz on 31 May 2013 at 00:00:00–01:00:00 UTC. The two frequency bands were chosen to demonstrate the behavior of the polarization analysis for two differing scenarios. At f_1 the Capon beamformer estimates the maximal energy arriving at the array with a back azimuth around $\sim 180^\circ$ and some weaker energy from other southern back azimuths, Figure 2.6a. For the polarization analysis of the same data, a histogram in Figure 2.6b of Rayleigh wave arrivals estimated with the method by Schimmel et al. (2011) is shown. The polarization analysis was used on all 13 stations of the array and only Rayleigh waves which exceed the degree of polarization (DOP) threshold of 0.75 were used for the generation of this figure. There is a good agreement between the back azimuth of the beam power spectrum and the polarization analysis despite the fact that multiple sources are contributing to the observed wavefield. The success can be attributed to the approach via the time-frequency domain, where the DOP constraint is used as a quality control, which allows to pick out short time periods where the polarization can be accurately estimated. For the approaches which average over longer time windows in time and frequency domain (Park et al., 1987; Jurkevics, 1988), the estimated back azimuths of each station show a larger scatter and are predominantly within the bounds of $180^\circ - 250^\circ$ (not shown). The higher scatter can be attributed to the fact that one computes the covariance or cross spectral density matrix from a window with multiple polarization states which can lead to an eigen basis which is not representable of the strongest polarization direction.

At the higher frequency f_2 , the Capon beamformer estimates sources from a wide range of back azimuths (Figure 2.6c). The polarization analysis estimates most of the Rayleigh waves from the west, and does not represent the overall state of the wavefield. The DOP constraint, while ensuring robustness in the result, prevents us from observing the complete source distribution in this example, and furthermore leads to relatively few accepted polarization measurements for this

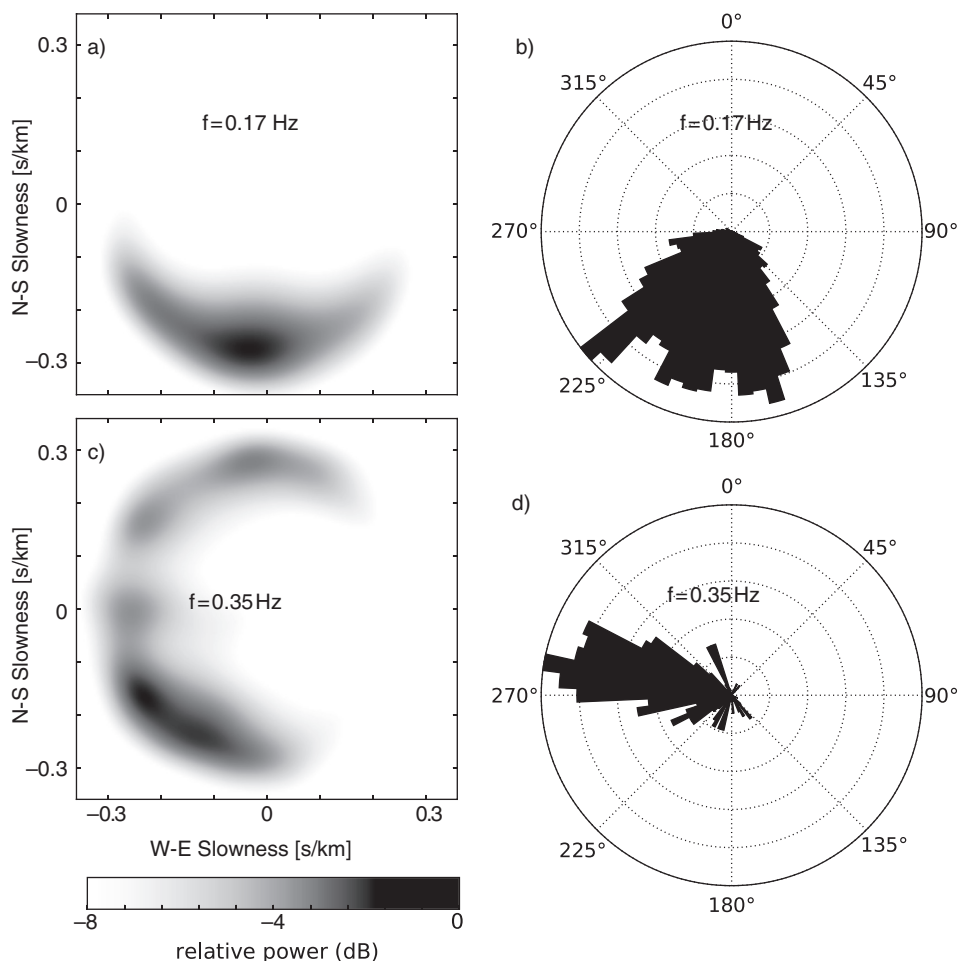


Figure 2.6. Results from (a,c) beamforming and (b,d) polarization analysis in two differing frequency bands for one hour of the ambient noise wavefield at PSAR (Australia). The first column shows the Capon power spectrum estimated from the vertical component and the left column shows a Rayleigh wave occurrence histogram. The radial axis on the histograms was normalized.

particular hour. If the DOP threshold is lowered to 0.5, the strong sources to the south-west are observed, although one can expect higher bias due to mixed polarization states. Hence, one should keep in mind that the DOP filtering procedure does not ensure the observation of the strongest source.

The window averaging techniques (Park et al., 1987; Jurkevics, 1988) estimate a back azimuth between 220° and 250° on approximately half the stations, while the other half shows large scatter and hence would benefit from a larger sample size of observations to increase the confidence in the results (not shown).

The second example with the Gräfenberg Array details observations for three differing frequency bands centered on $f_1 = 0.067 \pm 0.003$ Hz, $f_2 = 0.133 \pm 0.003$ Hz, and $f_3 = 0.25 \pm 0.01$ on 1 January 2017 at 01:00:00-02:00:00 UTC. The results from the f-k analysis are displayed in Figure 2.7a–c. The same data was previously used to discuss deconvolution techniques (Figure 2.3). F-k beamforming gives

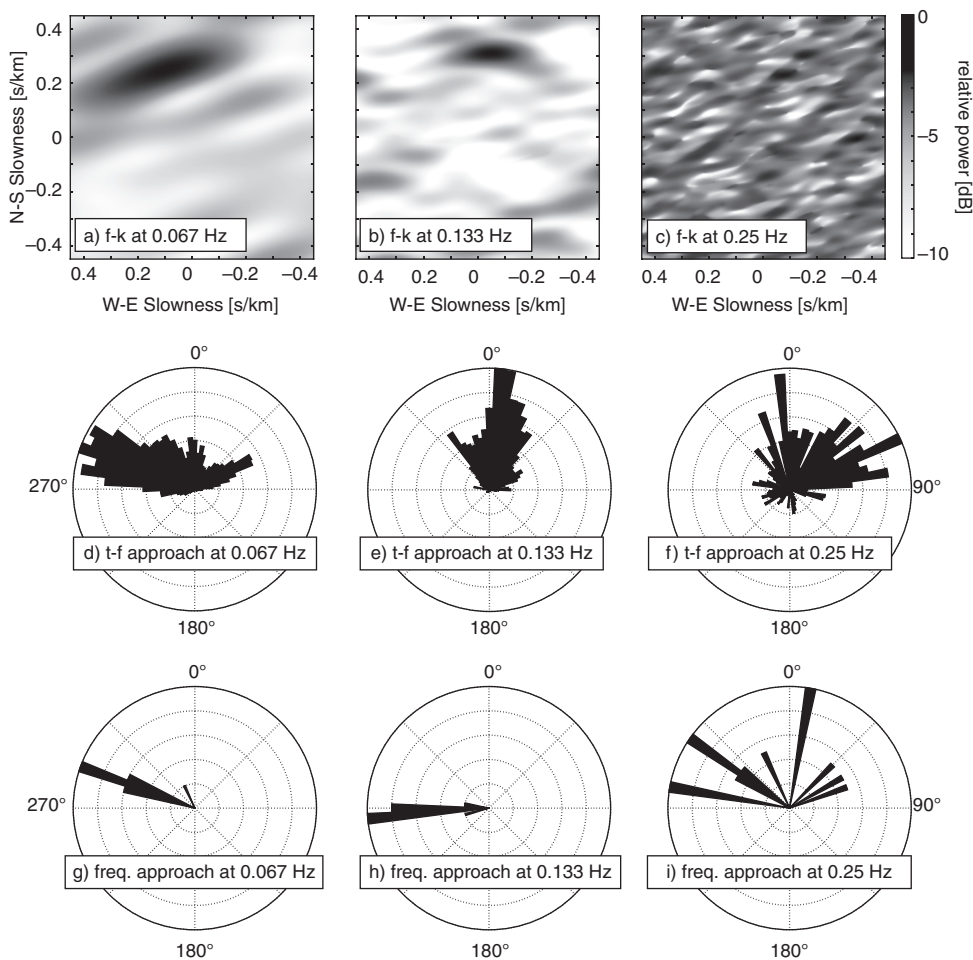


Figure 2.7. Comparison between results from f-k beamforming, time-frequency, and frequency polarization analysis. The top row (a-c) shows f-k beam power results at three differing frequencies. The middle row (d-f) displays the time-frequency approach and the bottom row (g-i) the time averaged frequency approach. The polar histogram plots display all accepted polarization directions above a set DOP for the time-frequency approach, while for the frequency approach the histogram is simply one estimated polarization direction per station.

reasonable results in the first two frequency bands but fails to give a meaningful estimate at 0.25 Hz. This is due to the small number of stations and the relatively large (100 km) aperture of the Gräfenberg Array. The approach is not able to identify plane wave propagation, and hence it is likely that path propagation effects are non-negligible.

The time-frequency polarization approach (Figure 2.7d–f), with a DOP threshold of 0.9, 0.8, and 0.6 for f_1 , f_2 , and f_3 respectively, shows partial agreement with the f-k beam power results. At f_1 , the polarization approach estimates the source location around 300° which is roughly 30° off from the f-k result, but coincides with the source distribution found by the Capon beamformer or the CLEAN algorithm (Figure 2.3). Additionally, arrivals from the north-east are estimated. Looking at the data in more detail, it is found that only 4 stations observe these signals from the north-east and it is likely a local source. Given that it does not seem to propagate as a plane wave, the f-k analysis does not indicate signals at the corresponding back azimuth. At f_2 good agreement with the f-k analysis is found and the main source is well estimated. At f_3 the f-k analysis fails, but the polarization analysis can still yield directional information.

The frequency polarization approach (Park et al., 1987; Jurkevics, 1988; Koper and Hawley, 2010) is displayed in Figure 2.7g–i. Given that the approach averages over time, we obtain only a single back azimuth value per station per hour, and hence the statistics and robustness of the approach are limited. At f_1 , an agreement with the time-frequency approach is found, and the frequency approach shows a much smaller spread. At f_2 , the results do not agree with the f-k analysis or the time-frequency approach, even though all stations seem to estimate the same back azimuth. A possibility is the mixing of multiple polarizations from differing back azimuths, and hence this approach would require longer time durations to obtain a statistically significant observation. Alternatively, the polarization direction would agree with a Love waves source to the north (confirmed with three-component beamforming). At f_3 the directions of polarization are distributed over a wide range of azimuths and the same logic with longer duration studies applies.

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